

Binary Search Notes

Binary search is an efficient algorithm for finding an item from a sorted list of items. It works by repeatedly dividing the search interval in half.

If the value of the search key is less than the item in the middle of the interval, narrow the interval to the lower half. Otherwise narrow it to the upper half.

The time complexity of binary search is $O(\log n)$, which makes it much faster than linear search for large datasets.

Binary search can be implemented recursively or iteratively using two pointers, typically named low and high.